DAILYN DESPRADEL RUMALDO

UX Researcher — Human Factors Researcher — Human-Centered Design — XR Interaction

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Summary

Doctoral UX and Human Factors Researcher specializing in understanding user needs and translating research findings into human-centered design solutions. Experienced in conducting user studies, evaluating usability, ergonomics, accessibility, and interaction performance to inform product and interaction design. Combining experimental research, physiological and behavioral analysis, and interactive prototyping, with industry experience at Apple and Meta and a PhD focus in XR and human-computer interaction.

Education

Carnegie Mellon University (CMU), Pittsburgh, PA **GPA: 3.77/4.00**
Ph.D. in Mechanical Engineering,Human-Computer Interaction Focus Anticipated Graduation: January 2027
Dissertation: Toward Adaptive and Accessible Extended Reality Interactions: Understanding and Enhancing User Experience

New York Institute of Technology (NYIT), New York City, NY **GPA: 3.66/4.00**
M.S. in Electrical and Computer Engineering Graduation: May 2021

Mercy College, Dobbs Ferry, NY **GPA: 3.71/4.00**
B.S. in Biology — *McNair Scholar* Graduation: May 2018

Industry Experience

Research Scientist Intern, Meta Inc. **October 2025 - March 2026**
User Experience and Robotics Systems

- Designed and conducted a user study with 18 participants to evaluate sEMG-based robotic teleoperation, analyzing task performance and user preferences to identify interaction opportunities and limitations
- Developed XR-based interaction prototypes integrating wearable sensing, force estimation, and robotic control to investigate alternative approaches to intuitive teleoperation
- Collaborated with five cross-functional teams to integrate research and system development, communicating findings and design implications to multidisciplinary stakeholders

Vision Pro Group (VPG) Product Design Intern, Apple Inc. **April 2025 - September 2025**
Ergonomics / Product Design Analyst Intern

- Conducted ergonomic evaluations and competitive benchmarking of XR hardware, using user studies, optical capture, and 3D scanning to assess comfort, fit, placement, wearability, and usability across diverse users.
- Analyzed large-scale 3D scan data, ergonomic measurements, survey responses, and existing user-research datasets using Python and Siemens NX to identify trends and deliver evidence-based product design recommendations considered in the next hardware review cycle.
- Led an independent human factors research project by defining objectives, assembling and coordinating a cross-functional team, and presenting evidence-based findings that informed product design and engineering decisions.

PhD Research Experience

Research Lead — Human–Computer Interaction, Accessibility & Extended Reality **August 2021 – Present**
NeuroMechatronics Lab, CMU (PI: Dr. Douglas Weber)

Research Focus: Adaptive XR Interactions: Understanding and Enhancing User Experience

Neuromechanics of Object Manipulation Across Physical and Virtual Environments 2021 – 2022

- Designed and conducted human-subject studies with 6 participants (5 male, 1 female) comparing object manipulation across physical, virtual, and haptic-enabled environments.
- Developed experimental and data analysis pipelines integrating EMG, motion capture, and hand kinematics to quantify movement behavior, muscle activity, and interaction performance.
- Analyzed user interaction strategies and haptic feedback effects to identify differences in physical and virtual manipulation, informing VR interaction evaluation and design.

Accessible Human–Computer Interaction Using Wearable sEMG 2022 – Present
Carnegie Mellon University x Meta Collaboration

- Led a collaborative CMU–Meta research program developing Unity-based interaction environments that enabled people with tetraplegia to control computer and XR applications using wearable surface electromyography (sEMG), resulting in a first-author manuscript under review at *Science Robotics*.

- Designed and conducted longitudinal usability and interaction studies with 7 participants (6 male, 1 female) with varying cervical spinal cord injuries, integrating task performance metrics, usability questionnaires, and participant feedback to evaluate accessible computer and XR interactions.
- Led an extended XR mobility and target-selection study with 25 participants (21 able-bodied, 4 with SCI), combining head/eye tracking, behavioral analysis, and user questionnaires to assess mobility limitations, interaction effort, comfort, and accessibility needs.

Adaptive, Force-Aware XR Interfaces for Accessibility *2025 – Present*

- Designed and conducted a preliminary study with 27 participants (20 male, 7 female) to investigate wearable sEMG-based force estimation and its application to continuous, force-responsive XR interaction
- Developed machine learning pipelines to estimate continuous hand force from physiological signals, enabling real-time force-based manipulation in immersive XR.
- Evaluated an XR panel manipulation task with 21 participants (11 male, 10 female), where users adjusted a virtual panel's configuration to match a target reference, incorporating and collecting performance metrics and questionnaire responses to evaluate the interaction experience

Selected XR Interaction & Prototyping Projects

Clean the Dishes VR | *Unity, Eye/Head Tracking, Controllers, EMG* **Repository**

- Designed and developed a multimodal VR interaction prototype supporting gaze, head movement, controllers, and EMG-based input for object selection and manipulation, exploring alternative interaction methods for accessible XR.

Peg-and-Hole Force Task | *Unity, Force Input, Hand Tracking, Data Logging* **Repository**

- Developed a force-sensitive VR manipulation task integrating randomized configurations, force constraints, hand-joint tracking, and real-time performance logging to investigate motor control and interaction behavior.

How Do You See the World? | *Unity, VR Accessibility, Post-Processing* **Repository**

- Created an immersive VR experience simulating visual impairments, including cataracts, glaucoma, and color-vision deficiencies, to demonstrate accessibility challenges in virtual environments.

Ultraleap Mid-Air Interaction | *Unity, Ultraleap, Hand Tracking* **Repository**

- Designed and prototyped controller-free desktop interactions using hand tracking and mid-air gestures, exploring alternative input techniques for object manipulation.

Inclusive Hand Animation Library | *Blender, Modeling, Rigging, Animation*

- Developed diverse, rigged 3D hand models and animations to support inclusive XR prototyping, accessibility research, and interaction design.

Master's Research Experience

Research Lead - VR and EMG-Based Interaction for Prosthetic Rehabilitation **August 2019 – May 2021**
NYIT (Advisor: Dr. N. Sertac Artan)

- Designed and conducted an exploratory study with 2 female participants, developing a VR rehabilitation prototype integrating Oculus Quest 2 and Myo Armband to evaluate EMG-based gesture control for virtual arm manipulation.
- Analyzed gesture recognition, system latency, and interaction responsiveness to identify technical limitations and inform the usability and design of assistive XR interfaces.

Technical Expertise

UX Research & Human Factors: Human-Subjects Research, Experimental Design, Usability Evaluation, Ergonomics, Accessibility, Behavioral Analysis, Quantitative User Research
Machine Learning & Data Analysis: Python, Machine Learning (MLPs, Transfer Learning), Signal Processing, Statistical Analysis, Sensor Fusion, Data Visualization
XR & Interaction Development: Unity, Unreal Engine, C#, Unreal Blueprints, AR/VR/MR, Multimodal Interaction Design, Real-Time Prototyping
Human Sensing & Measurement: Eye Tracking, Head Tracking, Hand Tracking, EMG, EEG, Motion Capture, Force Sensing, 3D Scanning
3D Design & Prototyping: Blender, Siemens NX, Fusion 360, SolidWorks, 3D Scanning, Optical Measurement, Animation, Rigging
Languages: Spanish (Native), English (Professional Fluency)

Fellowships

- GEM Fellow, CMU, Mechanical Engineering, 2021 - Present
- McNair Scholar, Mercy College, Biology, 2018

Selected Publications and Academic Output

- **D. Despradel**, et al. "A Wearable Myoelectric Sensor Enables Human–Machine Interaction for People with Tetraplegia." *Science Robotics*, Under Review.
- **D. Despradel**, et al. "Enabling Advanced Interactions through Closed-Loop Control of Motor Unit Activity After Tetraplegia." *ACM Symposium on User Interface Software and Technology (UIST)*, Pittsburgh, PA, 2024.
- **D. Despradel**, et al. "Closed-Loop Control of Motor Unit Activity After Tetraplegia for Human-Computer Interaction." *Society for Neuroscience Annual Meeting*, Chicago, IL, 2024.
- **D. Despradel**, et al. "Real-Time Detection and Decoding of Motor Unit Activity Using a Wearable Neuromotor Interface in a Person with Motor Complete Tetraplegia." *Neural Control of Movement*, Dubrovnik, Croatia, 2024.

Additional presentations available upon request.

Leadership, Outreach & Lab Development

- **Social Media Manager, NeuroMechatronics Laboratory** — Created research highlights, outreach content, and student spotlights to increase the visibility of the lab's research.
- **Motion-Tracking Laboratory Development** — Led the design and development of a dedicated motion-tracking research space supporting biomechanics, XR, and human-subject experiments.
- **Black History Month VR Event (CMU)** — Led an immersive VR experience (*MLK: Now Is the Time*) for the College of Engineering DEI program, 2024.
- **Biomechanics Day Outreach** — Led K–12 STEM workshops on biomechanics, Blender animation, and human movement, 2023–2024.
- **Mentorship Chair, MEGSO** — Organized graduate student mentorship initiatives and community engagement, 2023–2024.